Zeno and Anti-Zeno Physics

Firas Abouzahr

Northwestern

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A “Naïve” Introduction to the Quantum Zeno Effect

- Consider a state initialized at t=0 as $|\Psi(0)\rangle = |e\rangle$ then the probability as a function of time to measure it in state $|e\rangle$ is:

$$P_{ee}(t) = \left|\langle e|e^{-i\hat{H}t/\hbar}|e\rangle\right|^2 \approx 1 - \langle V^2\rangle\frac{t^2}{\hbar^2} \qquad (1)$$

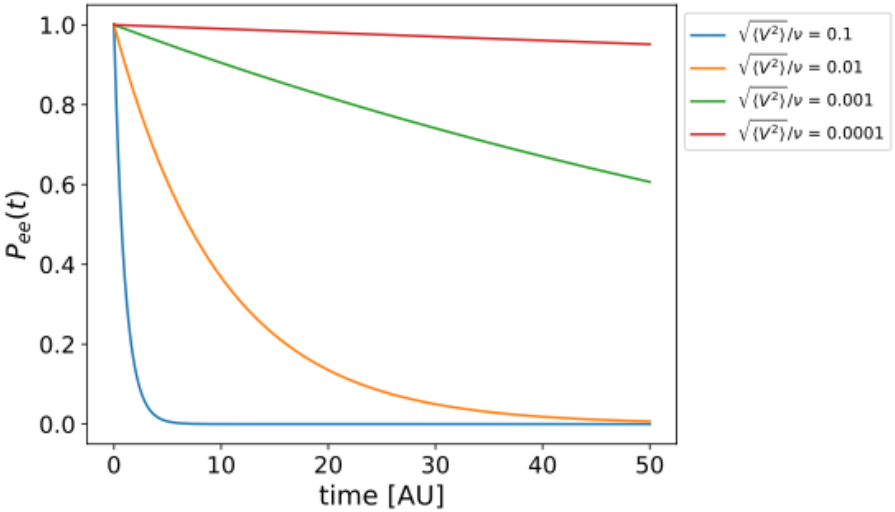
Expanding U around small t to $O(t^3)$

Kofman, A. G., & Kurizki, G. (2000). Acceleration of quantum decay processes by frequent observations. *Nature*, *405*(6786), 546-550.

- If we then ideally measure our system n times in intervals of $t = \tau$:

$$P_{ee}(t = n\tau) = \left(1 - \langle V^2\rangle\frac{\tau^2}{\hbar^2}\right)^n \approx e^{-\langle V^2\rangle t/(\nu\hbar^2)} \qquad (2)$$

Where $\upsilon \sim {}^1/_\tau$ is the measurement rate

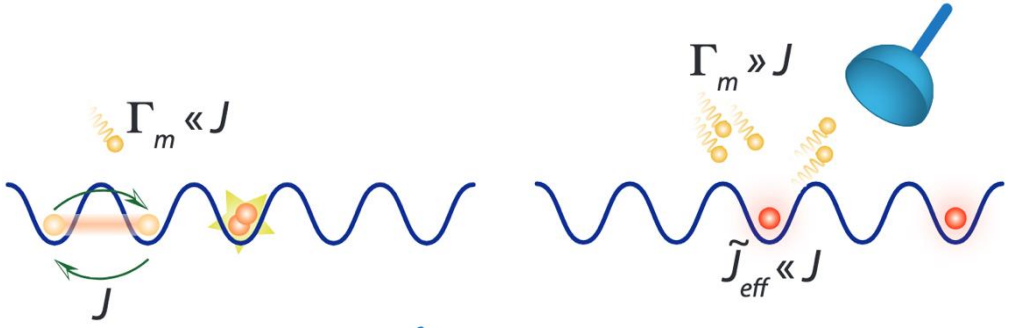


- With a high enough measurement rate, we repeatedly project our quantum state into measurement basis before it has time to coherently evolve into other states.
  - This can manifest as longer lifetimes in specific regimes!

This is the Quantum Zeno Effect (QZE)

How to Stop an Atom from Tunneling

- Patel et., al illustrate the QZE for ultracold gas confined to optical lattice sites.
  - The atoms in the gas tunnel across lattice sites with rate, J.
    - Due to this, atoms can then occupy same lattice site and then be lost via photo-association with a rate, κ.
  - The authors find with sufficiently large measurement rates, $\Gamma_m \gg J$, the atoms localize to their lattice site and hence the tunneling rate decreases → consequentially κ decreases.



- We can understand this experiment quite easily with our naïve picture!
  - We can write a simplistic two-lattice site Hamiltonian as: $\hat{H} \sim -J\left(\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}\right)$, one can then find that $\langle V^2 \rangle \sim J^2$
  - Hence, we find that $P_{aa}(t = n\tau) \approx e^{-\langle V^2 \rangle t/(\nu \hbar^2)} = e^{-J^2 t/(\Gamma_m \hbar^2)}$ and from last slide we expect to see the QZE when $\Gamma_m \gg J$.
    - Just as the authors found! **So why do do we need a better picture?**

Problems with the Naïve Picture

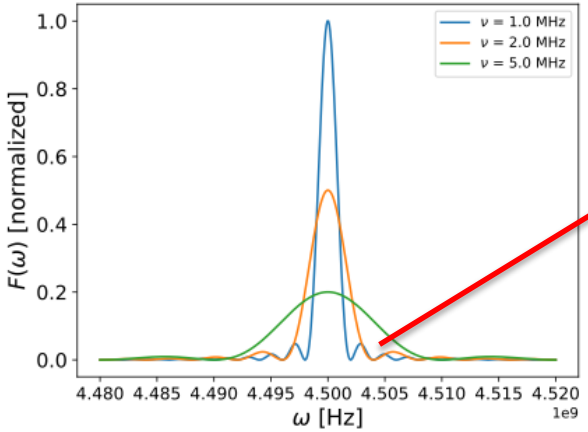
- In the "Naïve" picture, we assume ideal measurements. In reality, measurements are subject to the Energy-Time Uncertainty Principle:

$$\Delta E \Delta t \sim \hbar$$

Where $\Delta t \sim {}^{1}/_{\upsilon}$ such that as we increase υ, the spread in energy gets larger.

- This can cause our quantum state to couple to undesired channels.
  - This applies to systems like a radioactive particle that can decay into a continuum, or a qubit coupled to a bath

- Rather than considering ideal measurements, we can instead consider the transition spectral profile, $F(\omega)$.
  - $F(\omega)$ has a width on the order of υ and hence captures effects due to the Energy-Time Uncertainty Principle.



As we increases the measurement rate, the transition profile of our quantum system can start coupling stronger to channels away from its "bare" resonance.

Zeno vs Anti-Zeno Effects

- Now let's consider the following modified Hamiltonian to account for our system coupling to many possible decay channels:

$$\hat{H} = \hat{H} + \sum_j \left(V_{ej}|e\rangle\langle j| + V_{je}|j\rangle\langle e|\right) \quad \text{with wavefunction:} \quad |\Psi(t)\rangle = \alpha(t)e^{-i\omega_a t}|e\rangle + \sum_j \beta_j(t)e^{-i\omega_j t}|j\rangle$$

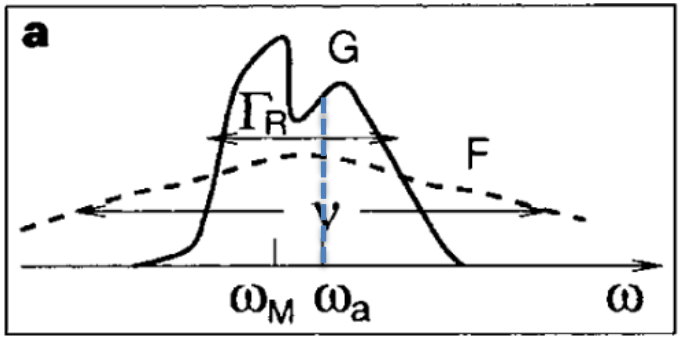
- By solving the Schrödinger and using some mathematical tricks, one can derive that $P_{ee}(t = n\tau)$ is given by:

$$P_{ee}(t = n\tau) \approx e^{-Rt} \qquad \text{where} \qquad R = 2\pi \int_0^\infty d\omega F(\omega) G(\omega)$$

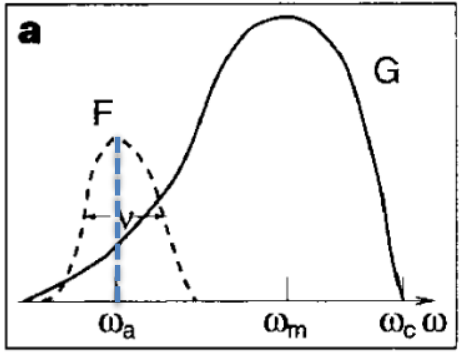
- $F(\omega) \equiv$ measurement broadened spectral transition profile
- $G(\omega) \equiv$ spectral density of states $|j\rangle$ weighted by their respective coupling strengths to $|e\rangle$

Kofman, A. G., & Kurizki, G. (2000). Acceleration of quantum decay processes by frequent observations. *Nature*, *405*(6786), 546-550.

Zeno Effect: $\nu \gg |\omega_a - \omega_m|, \Gamma_R$



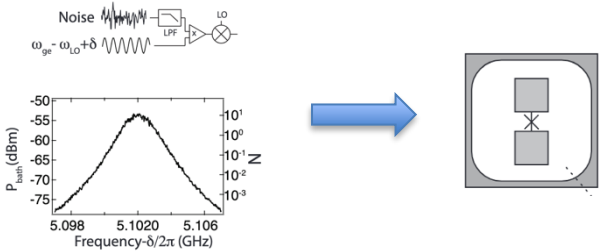
Anti- Zeno Effect: $\nu \ll |\omega_a - \omega_m|$



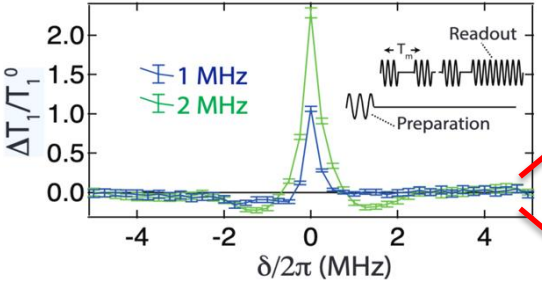
How to Make a Good Qubit Go Bad

- Harrington et., al illustrate the Zeno and anti-Zeno effects in sc qubits which can be understood via this overlap integral formalism.
  - The authors synthesize bath noise and control the detuning $\delta = \omega_q - \omega_m$.

Harrington, P. M., Monroe, J. T., & Murch, K. W. (2017). Quantum Zeno effects from measurement controlled qubit-bath interactions. *Physical review letters*, *118*(24), 240401.



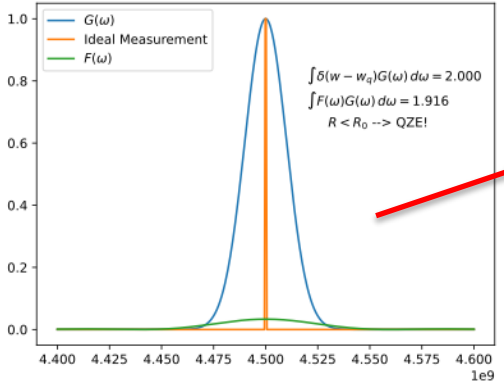
Couple superconducting qubit to engineered bath



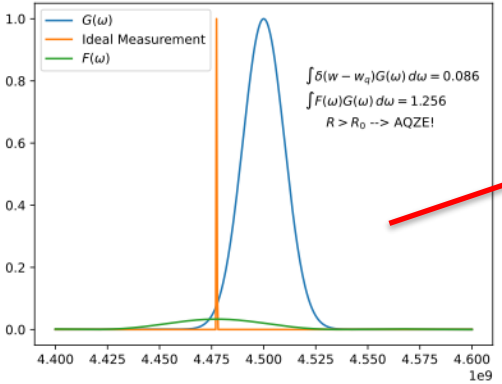
QZE (T_1 increases)

QAZE (T_1 decreases)

- We can pictorially understand the result on the right with overlap integrals:



When the qubit sits at the bath peak, measurement broadening lowers the average bath coupling, decreasing the decay rate (Zeno effect).



When the qubit is off-resonance from the bath peak, measurement broadening allows the qubit spectrum to overlap stronger regions of the bath, increasing the decay rate (anti-Zeno effect).

How to Numerically study the (A)QZE Effect

- The Quantum Zeno Effect is fundamentally a measurement phenomenon.
- We need a formalism(s) that describe quantum state evolution conditioned on measurement outcomes.

Stochastic Master Equation (SME)

Gambetta, J., Blais, A., Boissonneault, M., Houck, A. A., Schuster, D. I., & Girvin, S. M. (2008). Quantum trajectory approach to circuit QED: Quantum jumps and the Zeno effect. *Physical Review A—Atomic, Molecular, and Optical Physics*, *77*(1), 012112.

- SME leveraged to study the dynamics of our system under continuous observation of particular decay channels.

$$\dot{\rho}_{\bar{J}} = \mathcal{L}\rho_{\bar{J}} + \underbrace{\sqrt{\Gamma_{ci}}\mathcal{M}[\sigma_z]\rho_{\bar{J}}\left(\bar{J}(t) - \sqrt{\Gamma_{ci}}\langle\sigma_z\rangle_t\right)} - \underbrace{\frac{i\sqrt{\Gamma_{ba}}}{2}[\sigma_z, \rho_{\bar{J}}]\left(\bar{J}(t) - \sqrt{\Gamma_{ci}}\langle\sigma_z\rangle_t\right)}$$

ordinary Lindblad evolution

Updates the system towards $\pm\sigma_z$ eigenstates

Stochastic phase rotations about z

$\bar{J}(t) \equiv$ Measurement record. For sc circuits, homodyne detection of photons leaving the resonator gives: $\bar{J}(t) = \sqrt{\Gamma_{ci}}\langle\sigma_z\rangle + \xi(t)$ → Gaussian white-noise

$\rho_{\bar{J}} \equiv$ density matrix conditioned on $\bar{J}(t)$.

$\Gamma_{ci} \equiv$ rate at which we gain information about the qubit

$\Gamma_{ba} \equiv$ rate of measurement back-action

- Each realization of the stochastic measurement record $J(t)$ produces a different evolution $\rho_{\bar{J}}(t)$ → **we get individual quantum trajectories**
- $\rho(t) = E[\rho_{\bar{J}}(t)]$ → The density matrix from the standard master equation is the **ensemble average of all stochastic quantum trajectories**.
- **Zeno:** By numerically solving the SME with various values of Γ_{ci}, we can see how measurement rates give rise to Zeno effects.

How to Numerically study the (A)QZE Effect

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Monte-Carlo Wavefunction Method

Mølmer, K., Castin, Y., & Dalibard, J. (1993). Monte Carlo wave-function method in quantum optics. *Journal of the Optical Society of America B*, *10*(3), 524-538

- In the quantum jump (Monte-Carlo wavefunction) method, the system's wavefunction evolves under a non-Hermitian effective Hamiltonian and undergoes stochastic jumps generated by collapse operators:

$$\hat{H} = \hat{H}_0 - \frac{i\hbar}{2}\underbrace{\left(\sum_n \hat{C}_n^\dagger \hat{C}_n\right)}_{\text{Jump operators}}$$

$$\left|\phi^{(1)}(t+dt)\right\rangle = \left(1 - \frac{i}{\hbar}\hat{H}dt\right)|\phi(t)\rangle$$

Renormalize after each time step

- Each stochastic sequence of jumps produces a different wavefunction trajectory $|\phi_j(t)\rangle$
- With larger Γ_m, we cause our wavefunction to jump with higher likelihood → this can prevent coherent evolution, the Zeno effect!
- $\rho(t) = E\left[|\phi_j(t)\rangle\langle\phi_j(t)|\right]$ → The density matrix from the standard master equation is the **ensemble average of all stochastic quantum trajectories**.